FORMALIZING GAME LOGIC WITH SABOTAGE

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Abstract

Game logic with sabotage is an extension of Parikh's propositional game logic and is recently shown to be equiexpressive with the modal μ-calculus [1]. In this thesis, we provide a new formalization of game logic with sabotage and the modal μ-calculus, and prove their equiexpressiveness in the theorem prover Isabelle. The development has three stages, which represent successive aspects of contribution beyond the original mathematics work:

1. Two more general logics, recursive game logic and fixpoint logic with chop, are formalized and proven equiexpressive. A precise criterion for syntactic validity is defined and verified.

2. A generalized Bekič principle is precisely stated, proven mathematically and then formalized.

3. Game logic with sabotage and modal μ-calculus are realized as inductive subsets of recursive game logic and fixpoint logic with chop respectively and then proven equiexpressive.

This work establishes a formal link between game logic with sabotage and μ-calculus. One immediate application is that all model-theoretic properties transfer between the two logics for free. A more distant application is model-checking based on game logic, after proof calculi are formalized in future work.

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Contents

Chapter 1

Introduction

1.1 Notations and Conventions

1.1.1 Logic Names and Abbreviations

$\mathtt{GL}$	game logic
$\mathtt{GL_s}$	game logic with sabotage
$\mathtt{RGL}$	recursive game logic
$\mathtt{rlGL}$	right-linear game logic
$\mathtt{FLC}$	fixpoint logic with chop
$\mathtt{L}_\mu$	modal μ-calculus

1.2 Background

While fixpoint is known to be expressible in terms of iteration games, such understanding remains semantical. The syntactical objects served by these semantical games are fixpoint logics, including various forms of μ-calculi. This landscape was changed when $\mathtt{GL}$ was proposed by Parikh as a dynamic adversarial alternative for $\mathtt{L}_\mu$ [3]. In a sense, games are lifted from the semantical world to the syntactical world, and we can now express fixpoints in explicit iteration games.

The expressivity of $\mathtt{GL}$ in relation to $\mathtt{L}_\mu$ remained unclear until Wafa and Platzer extended $\mathtt{GL}$ in 2024 with the "sabotage" construct [1]. The resulting logic, $\mathtt{GL_s}$, has semantic-preserving two-way translations with $\mathtt{L}_\mu$ and the pair is therefore equiexpressive. This work illuminates the missing piece in $\mathtt{GL}$ that limits its expressive power, and hints at an excitingly new way of expressing formal specifications in terms of explicit *syntactical* games.

However, formalizations of either $\mathtt{L}_\mu$ or $\mathtt{GL}$ have not caught up with the mathematical development. $\mathtt{L}_\mu$ is a mature classical logic ubiquitous in modern model checkers, while $\mathtt{GL}$ has been there for decades. Hence, there is a need to formalize them both.

Another need for formalizing results in [1] comes from a recent refutation to an attempt at proving completeness of a proof calculus for `GL` [2]. If mathematical proofs can be shown to not work, then it makes sense to check them in a theorem prover.

1.3 Main results

We present the main results in the form of code snippets, accompanied by explanations. Most mathematical results are not new except the general Bekič principle. All formalizations are new.

Theorem 1.3.1. *In this thesis, we prove that*

$$e^{i\pi} + 1 = 0.$$

1.4 Structure of the thesis

In Chapter 2, we will make some necessary preparations. The proof of Theorem 1.3.1 will be given in Chapter 3.

Chapter 2

Preliminaries

2.1 Notation

Here is table of basic notations.

X	a compact metric space
$\mathscr{B}$	Borel σ-algebra
T	a continuous map from X to X
μ	a T-invariant Borel probability measure

Table 2.1: Table of notation

2.2 Definitions

Definition 2.2.1. Let X be a set. A *function* from X to Y is a mapping $f\colon X \to Y$.

2.3 Previous results

Theorem 2.3.1. *There are infinitely many primes.*

Chapter 3

Proof of main theorems

3.1 Lemmas

Lemma 3.1.1. *If $a \leq b$ and $b \leq a$, then $a = b$.*

3.2 Propositions

Proposition 3.2.1. *The relation $\leq$ is a partial order on $\mathbb{R}$.*

Proof. It is readily checked that $\leq$ is a preorder. Then the proof is completed by Lemma 3.1.1. $\square$

Chapter 4

Applications

4.1 Examples

Example 4.1.1. Let $A \subset \mathbb{R}^2$ be an self-affine set.

Figure 4.1: The Barnsley fern is a self-affine set[1].

[1] This figure is generated using a notebook from zfengg/PlotIFS.jl.

Appendix A

Glossary

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